

Kurzanleitung/Stichwortliste Laserspektroskopie

Der Versuch befindet sich im Aufbau. Es geht um eine Anordnung zur Messung von Absorptionsprofilen und zur dopplerfreien Laserspektroskopie an Rubidium. Zunächst muss der durchstimmbare Halbleiter-Dioden-Laser stabil auf der richtigen Wellenlänge betrieben werden. Ein Wavelengthmeter wird zur genauen Bestimmung der Wellenlänge genutzt.

Wenn der Laser stabil in dem gewünschten Wellenlängenfenster läuft, soll ein optischer Aufbau erstellt werden, mit dem doppler-verbreitete Absorptionslinien des Rubidiums aufgenommen werden können. Es soll untersucht werden, ob die Linien bei hoher Laserleistung in Sättigung gehen. Das Endziel des Versuches ist die dopplerfreie Aufnahme von Rb-Linien mit Hilfe der Sättigungsspektroskopie.

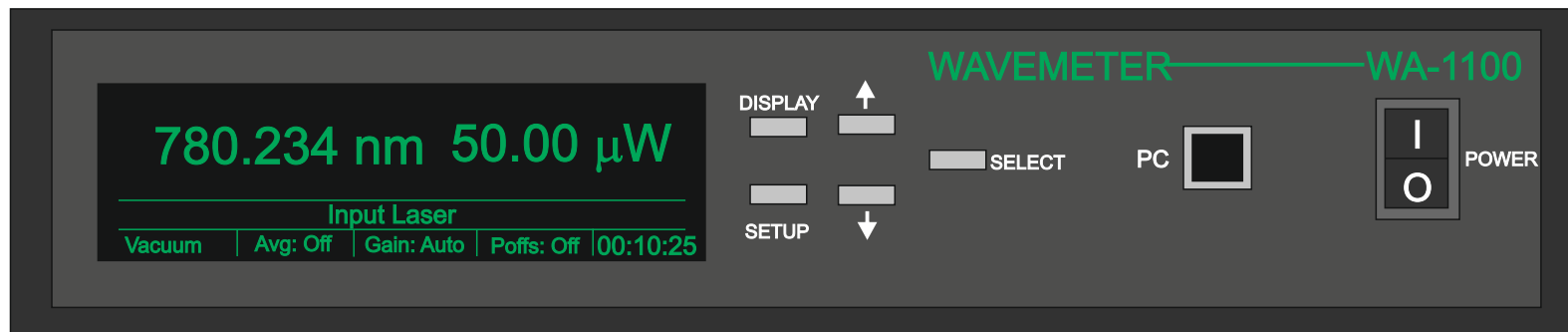
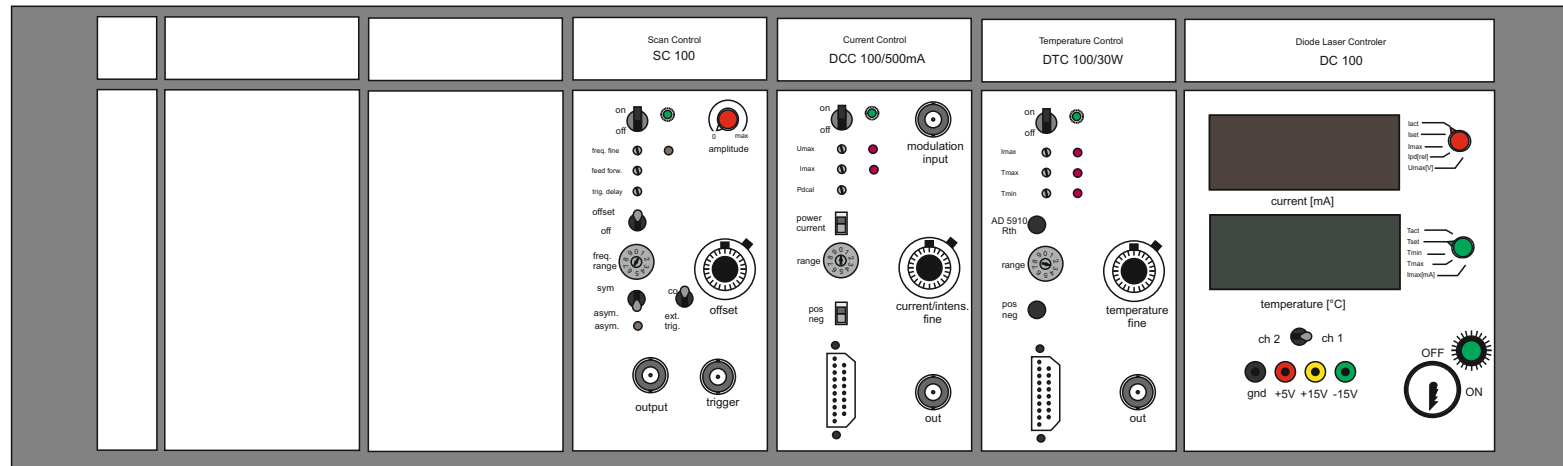
Folgende Punkte sind vorzubereiten, bzw. zu den entsprechenden Stichpunkten sollten Sie etwas sagen können:

- Grundlagen des Lasers, Einstein-Koeffizienten, Bestzung von Zuständen usw..
- Funktion einer Laserdiode, Stabilisierung des Lasers, Möglichkeit die Wellenlängen zu verändern, externe Moden im Littmann- und Littrow-Aufbau
- Auflösungsvermögen von Spektralapparaten
- Funktionsweise eine Wavelengthmeters
- Fabry-Perot , planar und konfokal
- Atomphysik der Rubidium-D-Linien (Feinstruktur, Hyperfeinstruktur)
- grundlegende Linienverbreiterung (nat. Linienbreite, Dopplereffekt, Sättigungsverbreiterung)
- Absorptionsspektren
- Dopplerfreie Spektroskopie, Lamb-Dip, Cross-Over, Grundprinzipien

Zur Vorbereitung steht zur Verfügung:

- Literatur Optisches Pumpen (Atomphysik Rb)
- Demtröder Laserspektroskopie Kapitel 7 (Laserspektroskopie)

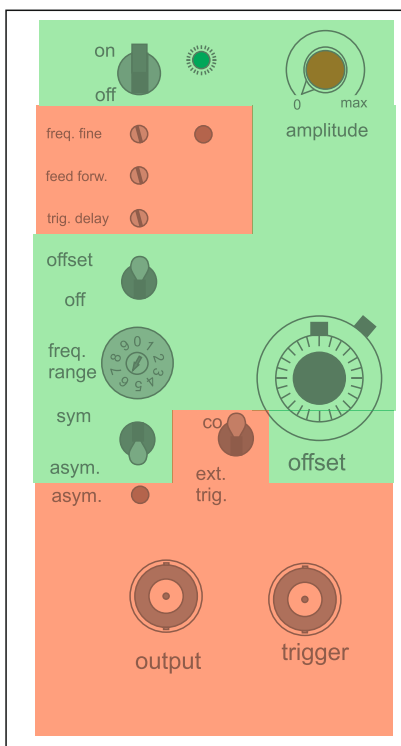
Laserspektroskopie



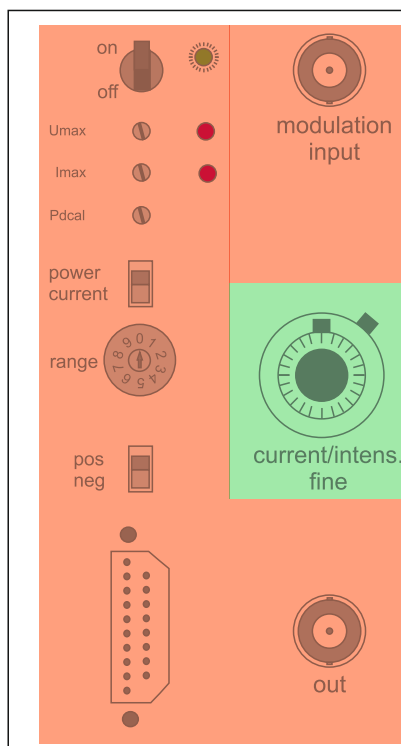
Do not touch these

You can touch these

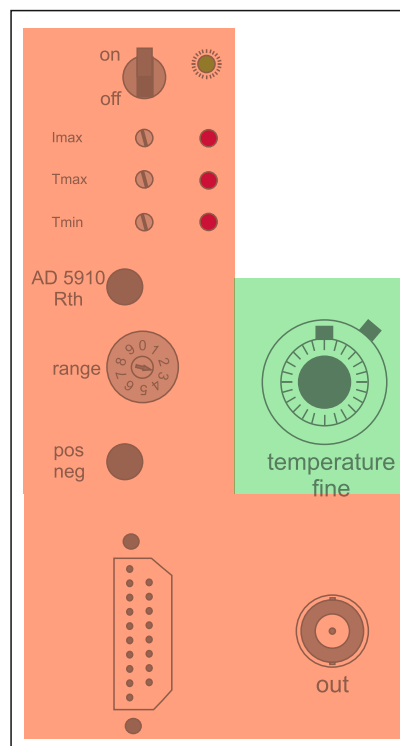
Scan Control
SC 100



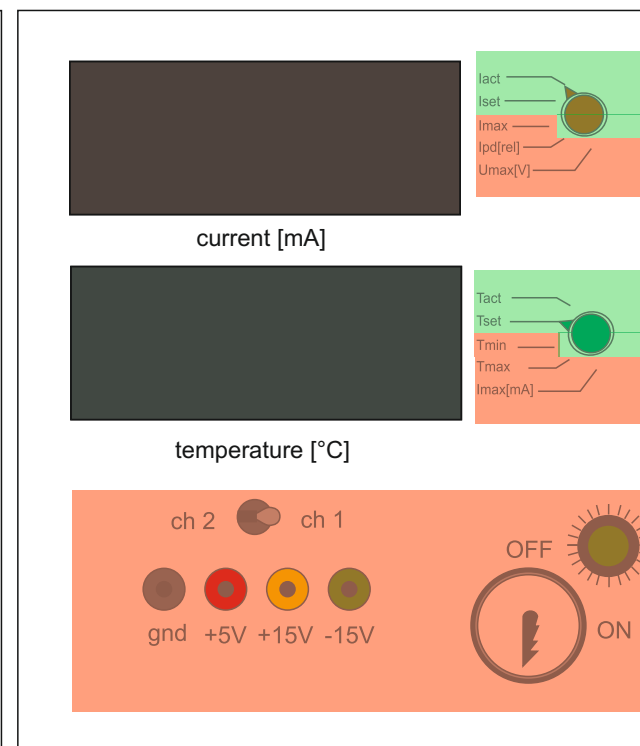
Current Control
DCC 100/500mA



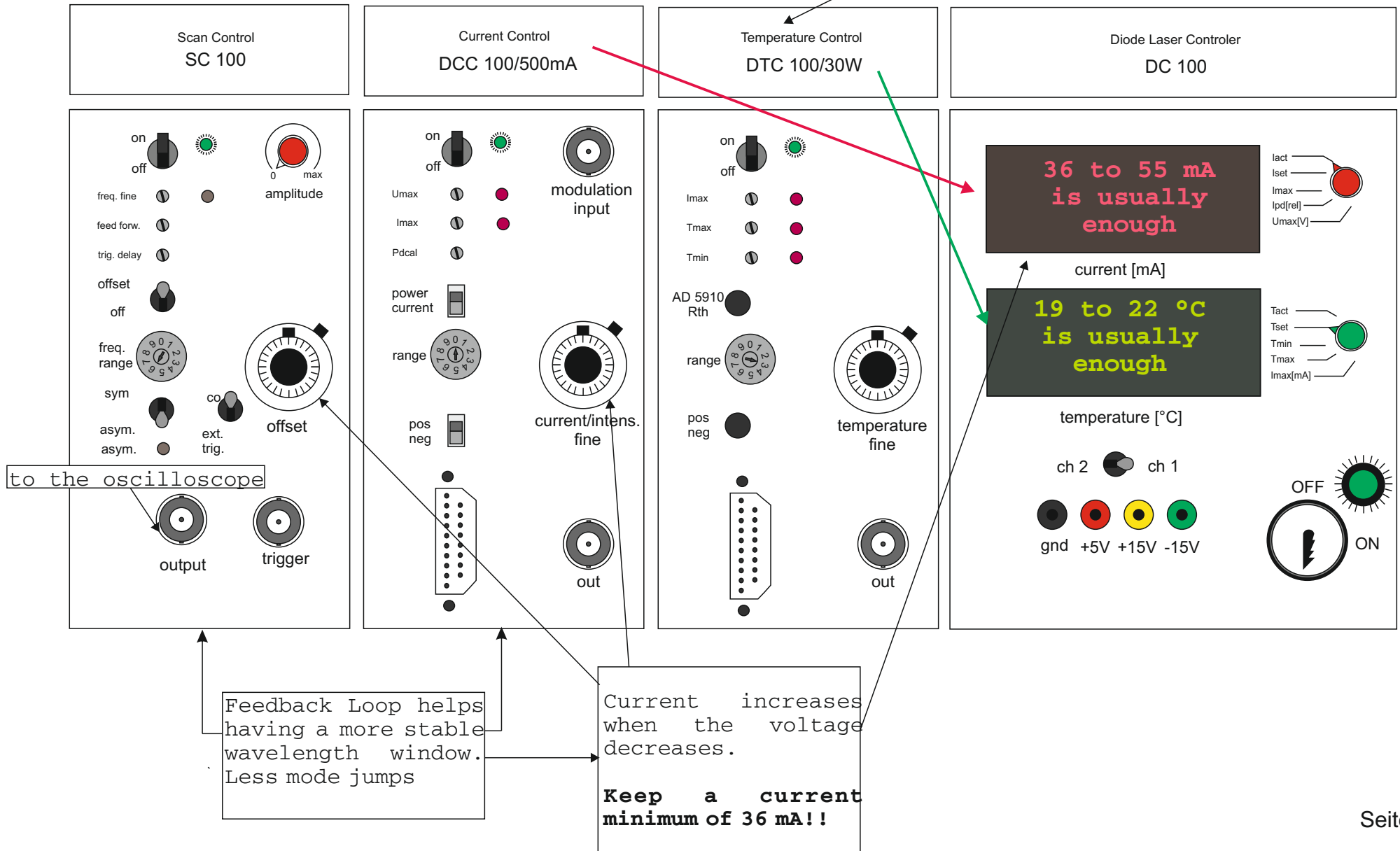
Temperature Control
DTC 100/30W



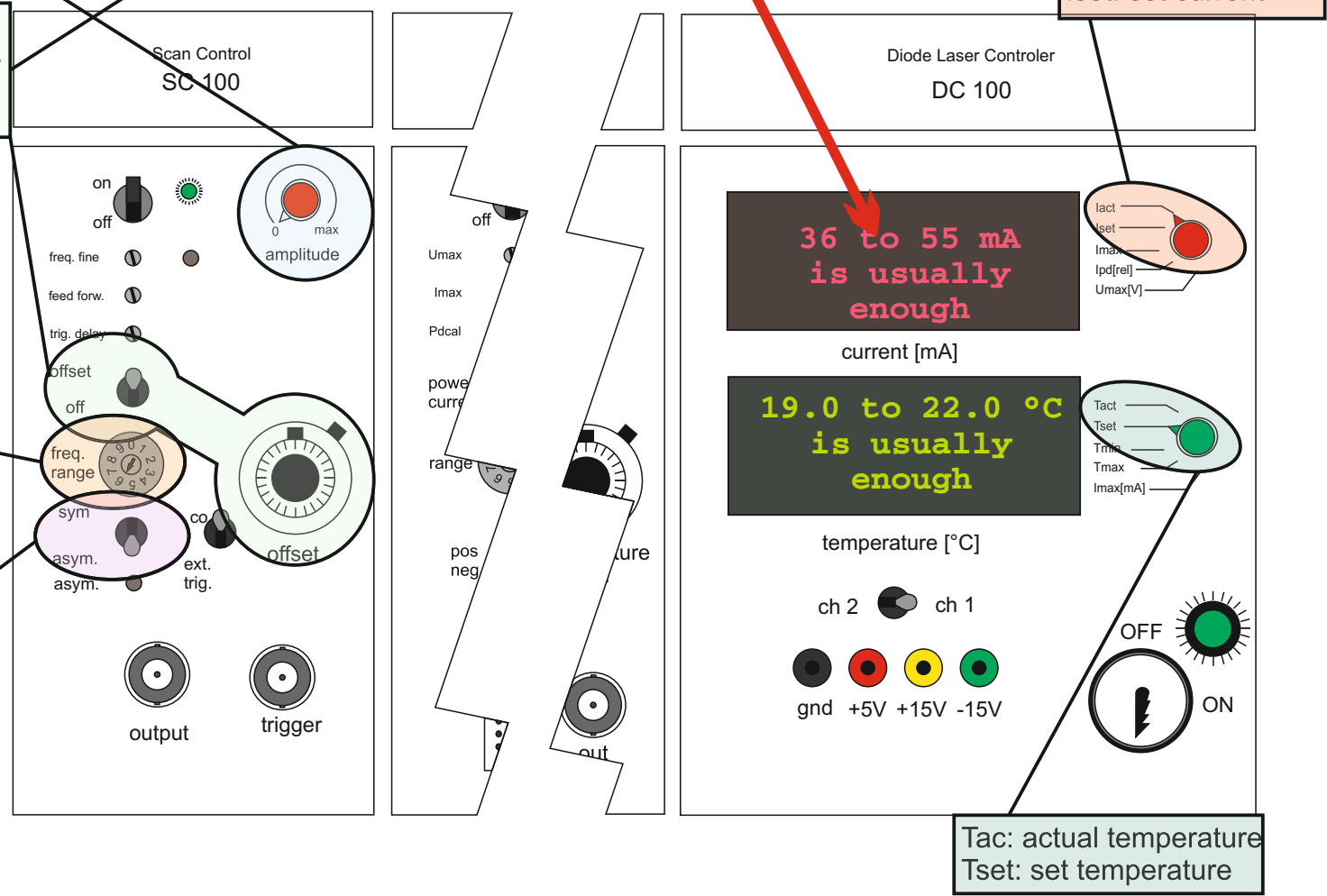
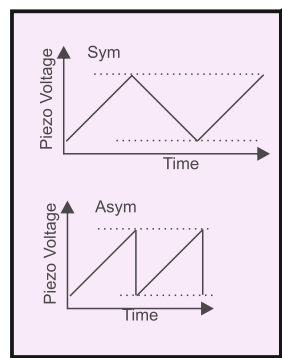
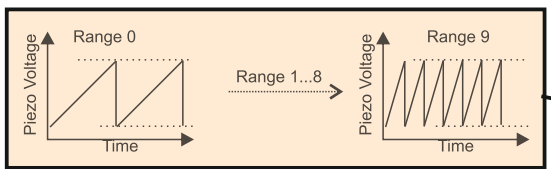
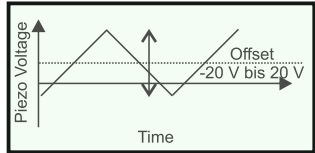
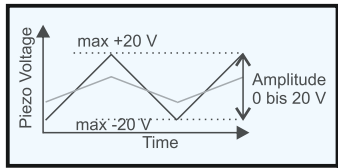
Diode Laser Controller
DC 100

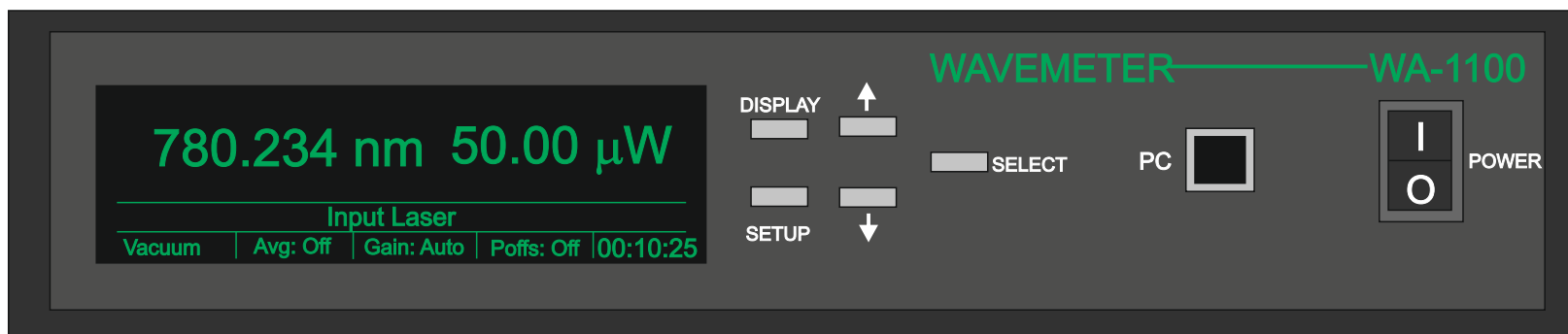


Temperature
stabilization may
take a very long
time if the setting
is changed by more
than 0.5 °C

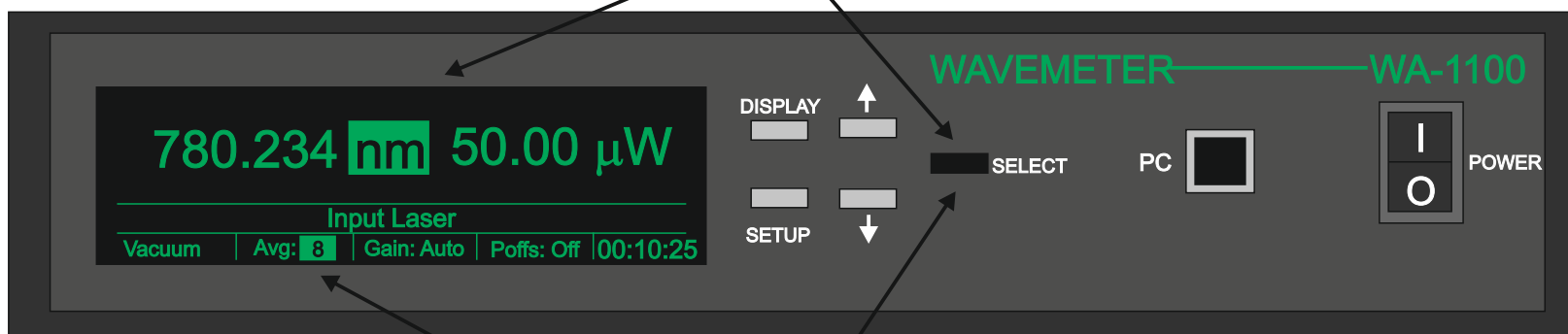


Warning!
Offset and amplitude change the current values (feedback-loop to stabilize the wavelength). Keep the current at a minimum value of 36 mA!





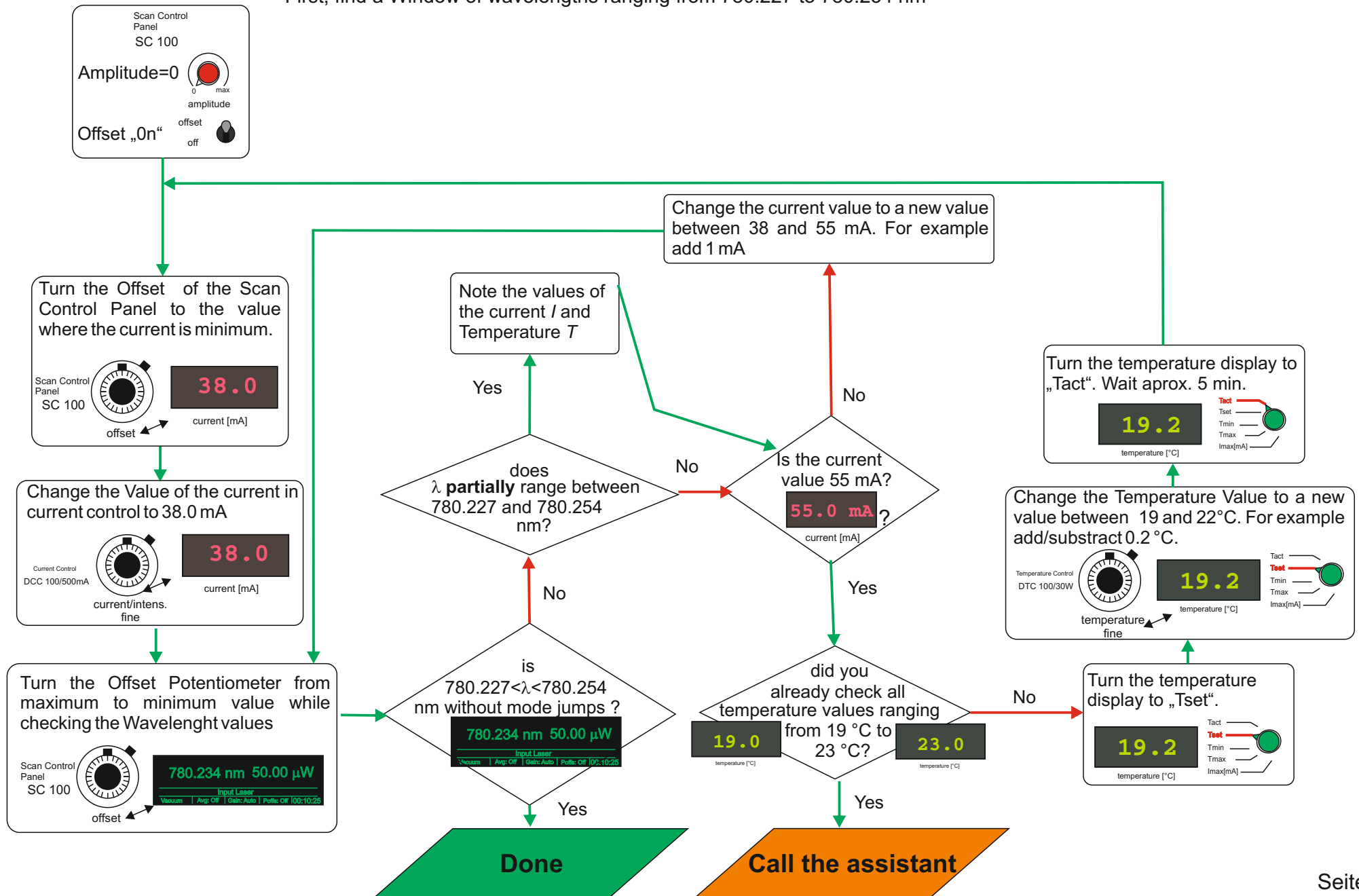
Press „select“ to change the displayed values of Wavelength or Frequency. Personally I prefer wavelength.



Press „select“ the number of averages. If the Piezo Voltage has Amplitude „0“ and is only offset-controlled, this may be helpful specially for calibration purposes

Task 1: Mode searching for the measurement

First, find a Window of wavelengths ranging from 780.227 to 780.254 nm



Task 2: Calibrating the Fabry-Perot

The Piezoelectric effect has some Frequency-Dependent Hysteresis and also creeping effect. A Calibration Wavelength Vs Piezovoltage does not make sense.

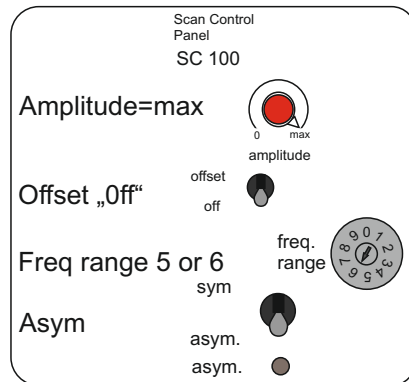
Task 2.1. Explain Piezoeffect, Hysteresis and Creeping effect.

Task 2.2. Explain the working Principle of a Fabry-Perot interferometer.

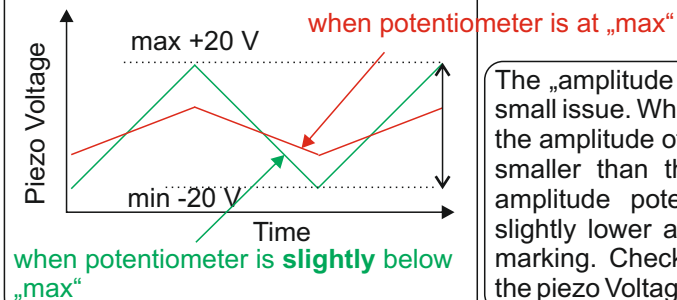
Task 2.3. How can we use a Fabry-Perot interferometer to calibrate the Wavelengths of our laser?

Task 2.4. Align the optical parts of the Laserspectroscopy experiment so that you have one photodiode passed the Rb cell and another photodiode passed the Fabry-Perot interferometer. Try to maximize the Voltage of the Photodiodes by improving the voltage. In order to do that you have to follow the next flow diagram.

Setup



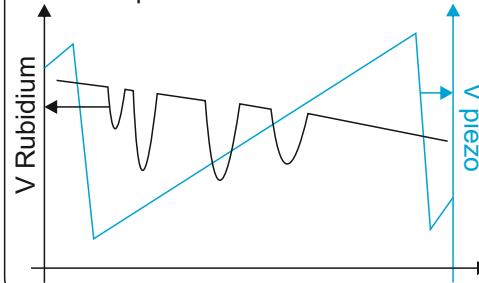
Maximize the Amplitude of the Piezo Voltage. Check at the oscilloscope.



The „amplitude potentiometer has a small issue. When turned to the max, the amplitude of the signal is slightly smaller than the signal when the amplitude potentiometer points a slightly lower angle than the „max“ marking. Check this by connecting the piezo Voltage to the oscilloscope.

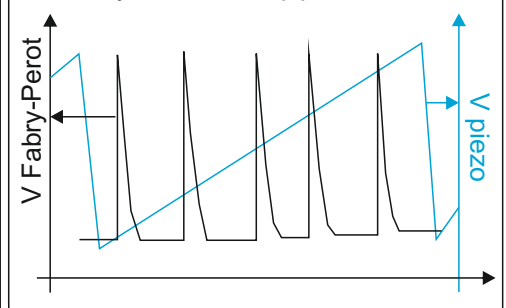
Align the Photodiodes 1.

-The Rb Cell photodiode usually saturates. Misalign it or cover it then a little bit. You want a signal on the oscilloscope like this:



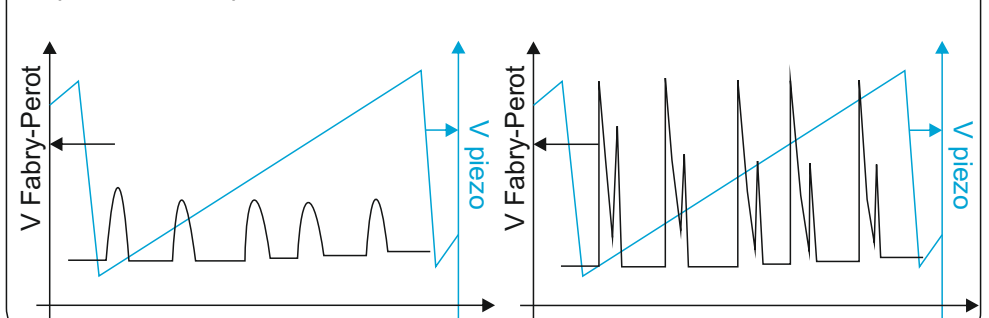
Align the Photodiodes 2.

-The Fabry-Perot photodiode is more challenging. You want a signal that shows 5 to 7 asymmetric sharp peaks.



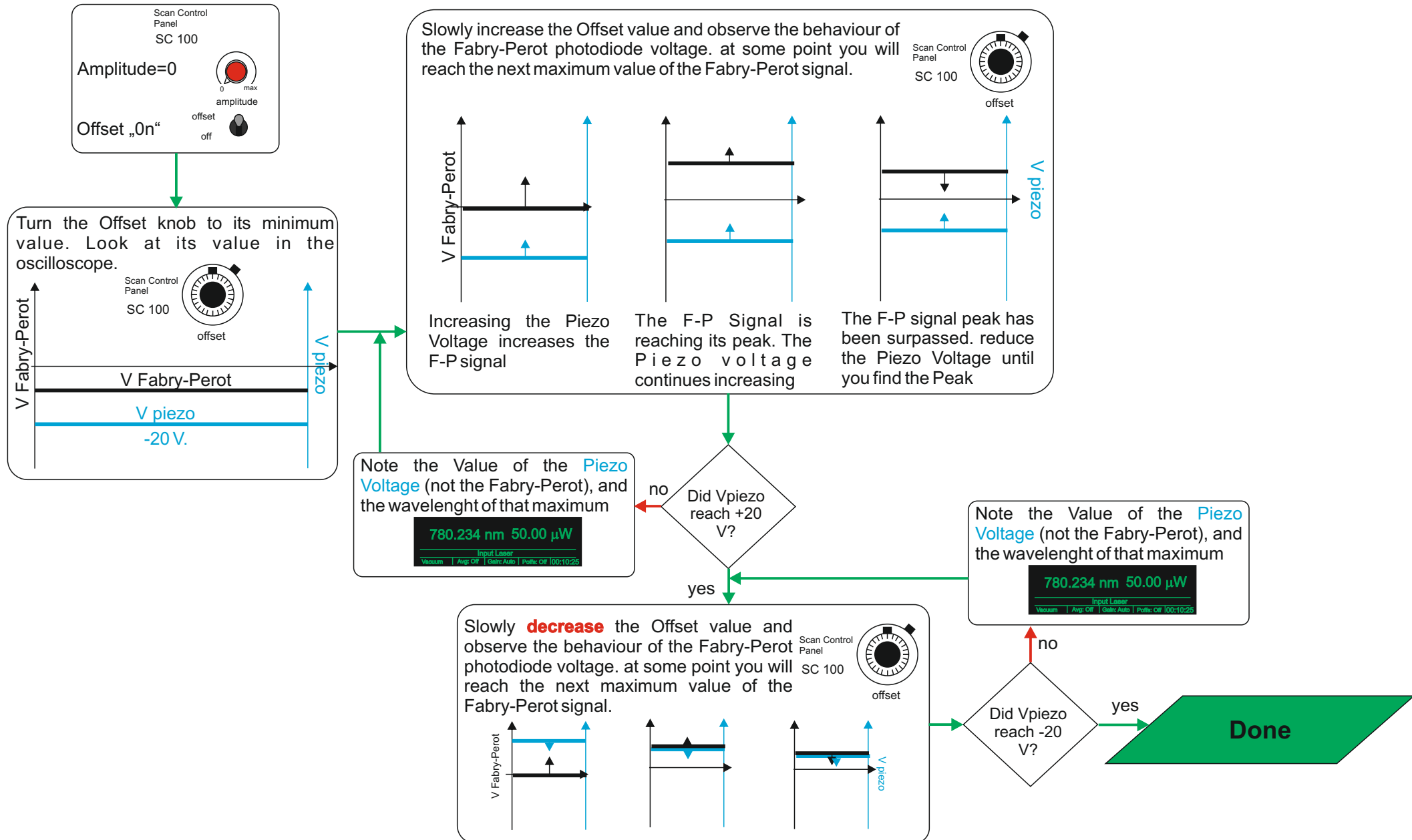
Align the Photodiodes 3.

-You may get signals like this from the Fabry-Perot. Align so that the signal looks like step 2 as much as possible.



Task 2: Calibrating the Fabry-Perot

Task 2.5. Calibrate the wavelengths at which the Fabry-Perot is transparent note the piezovoltage values in order to recognize these later during the measurement. Calibrate also the hysteresis in the positions of the wavelength.



Task 3: Taking measurements

Task 3.1: Build a setup first for doppler spectroscopy and take 3 Measurements for frequency number 1 and two other frequencies. Use the 4 channels measurement programm in the PC.

Task 3.2: Build a setup first for doppler free spectroscopy and take 3 Measurements for the same frequencies as before with a blocked pump beam and without it in order to subtract both signals. This task may not be succesful, in such case write a report for the doppler case. Should you be succesful dismiss Task 3.1 and write your report analysing only the doppler free measurement.

Task 3.3: in order to do the measurements take into account the following hints

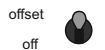
Adjust the piezo voltage amplitude and offset in order to see all of the Rb absorption peaks and avoid saturation signals in the piezo voltage.

Scan Control
Panel
SC 100

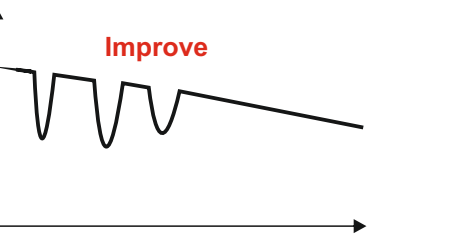
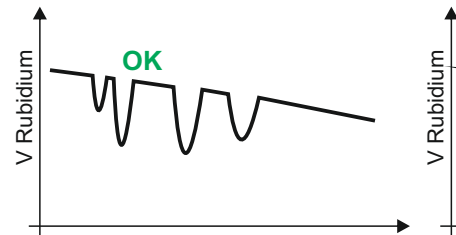
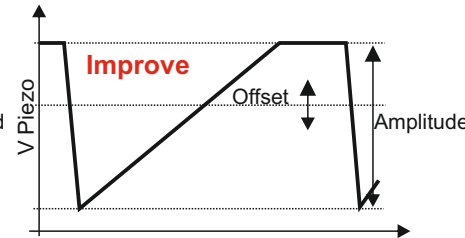
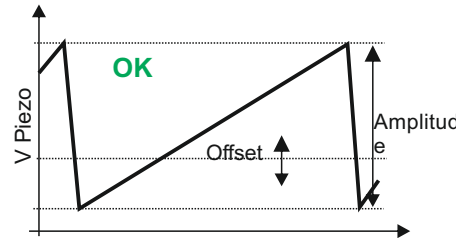
Amplitude $\neq 0$



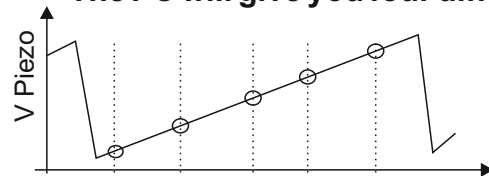
Offset „On“



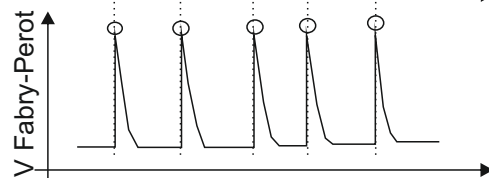
Freq range 5 or 6



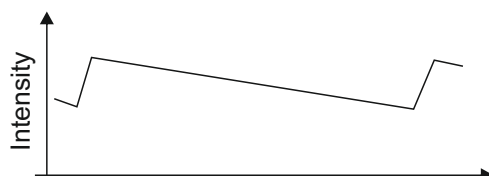
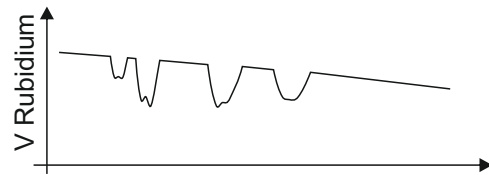
The PC will give you four different signals.



The Piezo Voltage can now be calibrated with the determined wavelengths of the Fabry-Perot signal.



You know the wavelengths corresponding to these maxima.



This signal shows the laser intensity coming at the Wavelengthmeter. Jumps in the intensity usually mean a jump in the wavelength mode.



Plot the Wavelengths Vs Piezovoltage for all three frequencies. You will see differences due to the hysteresis

Task 4: Data analysis and report

Task 4.1: Plot the Peaks of the Fabry-Perot Vs the Piezo voltage for all frequencies. You will see the effect of the hysteresis.

Task 4.2: Plot and fit all three calibration curves (V_{piezo} Vs Wavelength) for every frequency. Fit all three curves.

Task 4.3: Plot the Fabry-Perot Voltage Vs wavelength from the fits of all three curves to check that the fit corrects the hysteresis effects.

Task 4.4: Choose the best frequency to do the rest of the analysis of the Rb dopplerfree (or doppler) peaks. Explain why you have chosen that frequency.

Task 4.5: Why do we see only 4 absorption peaks? Demonstrate your answer with the measured data.

Task 4.6: Estimate the relative proportion of both Rb isotopes from your measurement.

Task 4.7: Explain your results thoroughly. Assign the corresponding electron transitions of the hyperfine structure to the measured data. Calculate relative deviations from literature values, try to explain them.